Akash Hadagali Persetti

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Summary

Applied AI engineer who ships agentic systems end to end, from LangGraph agent design through production AWS deployment. Built and shipped three live AI products: self-evaluating multi-agent workflows, a streaming LLM digital twin on Amazon Bedrock, and a voice-driven expense agent. Published an open-source MCP server to PyPI. Comfortable owning the full stack: agent logic, FastAPI REST backends, React and Next.js frontends, and Terraform-managed CI/CD on AWS. M.S. in Computer Science, Indiana University Bloomington (2026).

Projects

Wingman | *Self-Evaluating Agentic Co-Worker* | LangGraph, FastAPI, Lambda, DynamoDB wingman.akashpersetti.com

- Designed a worker-evaluator state machine in LangGraph where a structured-output evaluator checks worker responses against user-defined criteria and reinjects feedback for up to 5 retry turns, raising answer-validity rates before surfacing a result.
- Made the agent serverless-safe by reconstructing LangGraph state from DynamoDB on every request instead of an in-memory checkpointer, eliminating cold-start state loss across Lambda invocations.
- Integrated 5 tool domains through LangChain ToolNode and bind tools: web search (Serper), Wikipedia, sandboxed Python REPL, sandboxed file I/O, and Pushover notifications.
- Shipped the full stack on AWS with zero long-lived credentials: containerized FastAPI with Mangum, pushed to ECR, and provisioned Lambda, API Gateway v2, DynamoDB, and CloudFront through a Terraform-managed CI/CD pipeline using GitHub Actions OIDC.

Twin | *Streaming AI Digital Twin* | FastAPI, Next.js, AWS Bedrock (Claude Sonnet), S3, CloudFront akashpersetti.com

- Built a streaming personal-website agent grounded in a persona assembled at Lambda cold-start from a LinkedIn PDF, career summary, and style guide (parsed via pypdf) and injected into a Bedrock Claude Sonnet system prompt.
- Hardened against hallucination and prompt injection with a strict system prompt that forbids fabricated facts, blocks common jailbreak patterns, and constrains every response to verified persona context.
- Cut full-stack deployment to a single command across dev, test, and prod by wiring GitHub Actions and AWS OIDC to chain Docker build, Lambda upload, Terraform apply, Next.js export, S3 sync, and CloudFront invalidation.
- Persisted per-user conversation history at near-zero cost using session-keyed JSON in a private S3 bucket, with session IDs generated client-side through API Gateway.

TallyMark | *Voice + Text Agentic Expense Splitter* | React Native (Expo), FastAPI, NeonDB (PostgreSQL), LangGraph

- Engineered a dual-modal LangGraph agent that parses natural-language expense descriptions from voice or text, answers balance and group queries, and executes write actions (add expense, settle debt) against live user data via tool-bound FastAPI endpoints.
- Implemented a graph-based debt simplification algorithm that collapses N-way IOUs into the minimum-transaction settlement, toggled per group so users see either gross splits or net balances on demand, surfaced through a 4-tab React Native frontend over a modular FastAPI and NeonDB backend.

mcp-second-opinion | *Open-Source MCP Server, Published to PyPI* | Python pypi.org/project/mcp-second-opinion

- Built and published an MIT-licensed MCP server that lets any MCP-aware agent consult rival LLMs (OpenAI, Gemini, Anthropic, Grok) mid-conversation, exposing two tools: one to query a single model and one to fan out to all enabled providers in parallel and compare answers.
- Unified four provider APIs behind a single LiteLLM interface, returning per-call latency, token counts, and cost; gracefully disables providers with no API key instead of failing, and ships as a pip-installable CLI registered through standard MCP client config.

Experience

Machine Learning Intern (Remote) Jun 2025 – Aug 2025
MyEdMaster LLC *Leesburg, VA*

- Built a real-time computer-vision exercise-assessment system that cut repetition miscounting by roughly 30% and reached over 92% count accuracy across 3 exercise types (Kettlebell Front Raise, Seated Leg Extensions, Jack Knives) by adding calibration routines and motion-consistency checks in Python and MediaPipe, lowering estimated coach intervention time by about 40%.
- Held form-detection latency under 100ms per frame by engineering real-time pose-landmark extraction and joint-angle calculations, delivering sub-second corrective feedback so users could self-correct during live sessions without instructor oversight.

Web Development Intern (Remote) Oct 2023 – May 2024
Squadcast Labs *Bengaluru, India*

- Built 18+ responsive landing pages and 38 SEO-targeted blog pages and resolved Core Web Vitals bottlenecks to improve site performance and search visibility.

Education

Indiana University Bloomington Graduated May 2026
M.S. in Computer Science, GPA 3.8/4.0

The National Institute of Engineering, Mysuru, India 2020 – 2024
B.Eng. in Computer Science and Engineering, GPA 3.7/4.0

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, Java, Apex, SQL

Agentic AI & LLM Engineering: LangGraph, LangChain, OpenAI Agents SDK, CrewAI, AutoGen, Strands Agents, n8n, Model Context Protocol (MCP, published server to PyPI), function calling, structured outputs, prompt engineering, multi-agent workflows, LLM evaluation and observability

Cloud & DevOps: AWS (Lambda, Bedrock, S3, CloudFront, DynamoDB, ECR, API Gateway, EC2, SageMaker), GCP, Azure, Docker, Terraform (IaC), GitHub Actions CI/CD (OIDC), serverless and event-driven architecture

Generative AI & ML: RAG pipelines, ChromaDB (vector database), embedding models, Hugging Face, AWS Bedrock (Claude Sonnet), OpenAI, MediaPipe, CNNs, classification and clustering, scikit-learn

Backend, Frontend & Databases: FastAPI (REST APIs, microservices), Next.js, React, React Native (Expo), Node.js, PostgreSQL, NeonDB, MySQL, MongoDB, DynamoDB

Certifications

Proficient AI Engineer – Ed Donner / The AI Engineer track (2026) [verify] | **Introduction to Machine Learning** – NPTEL / IIT Madras (2023) [verify]